

Information-Aware Excitation Trajectories for In-Orbit Mass Property Estimation

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Abstract—When assembling a spacecraft on-orbit, the task of mass property estimation becomes important, since the mass properties of the entire structure cannot reasonably be measured prior to launch. Mass properties can be estimated in the microgravity environment by applying excitations with the spacecraft thrusters and measuring the resulting accelerations. However, such excitations disturb the structure from its equilibrium condition and consume scarce resources such as fuel and power. This work presents a method for information-aware trajectory planning, balancing information gathering on mass properties with reaching a desired goal state. We apply these methods to learn about the evolving mass properties of a compound superstructure after a new element is added during an on-orbit assembly scenario. Results demonstrate that mass and inertial estimation capabilities can be highly accurate ($< 1\%$ error), but are also highly sensitive to the underlying trajectory, require parameter-specific information tuning, and that online-parameter adaptation is a requirement for robust estimation over the course of a mission.

I. INTRODUCTION

Space missions are currently limited by the mass and volume constraints of available launch vehicles. Large structural elements of spacecraft are typically folded for launch and deployed in orbit. Recently launched missions like JWST are reaching the limit of what can be reasonably flown in a single launch. Recently, missions have been proposed with even larger structural elements, such as the NASA in-Space Assembled Telescope (iSAT) [6] and its 20 meter diameter mirror. On-orbit assembly offers an alternative way to deploy large space structures over multiple launches.

On-orbit assembly requires active attitude control to counter disturbances and prevent tumbling. Spacecraft attitude control techniques typically require some knowledge of mass properties, and spacecraft are often carefully designed and built so that their mass, moments of inertia, and center of mass are known prior to launch. These mass properties are also typically measured prior to launch to ensure they fall within the designed envelope. For a large space structure, measurement of the fully assembled spacecraft is not possible on the ground. Furthermore, these properties may shift dramatically over the course of the assembly process. For the task of attitude control during assembly, it is useful to be able to measure the mass properties of the spacecraft on-orbit so the attitude control algorithms can be designed based on up-to-date mass information.

Measurement of mass properties in the microgravity environment involves applying an excitation to the structure with onboard thrusters, and measuring the resulting accelerations to estimate the mass properties according to a dynamics model.

Such excitations must be applied intelligently so as to learn as much information about the mass properties as possible, while disturbing the spacecraft's equilibrium condition as little as possible, and consuming as little fuel as possible. To achieve these design requirements, we employ information-aware motion planning techniques to plan an excitation trajectory that gathers information about a trajectory during a spacecraft's normal maneuvers, such as slewing to point in a new direction.

This paper makes three main contributions. First, we develop and solve an optimal trajectory generation problem that balances mass property information gathering with other control objectives, such as reaching an equilibrium condition and minimizing control effort. Second, we propose an information weighting heuristic to decrease information weighting as the mass property estimates converge. Finally, we apply our method to an on-orbit assembly scenario to autonomously learn the new mass properties when a new element is attached to the structure.

II. PROBLEM FORMULATION

For the purposes of this study, we use a planar model of the microgravity system dynamics. We study a modular spacecraft consisting of active blocks, which have onboard thrusters, and passive blocks, which could be structural modules such as segments of a telescope mirror. We assume the spacecraft modules are attached rigidly. Applying the Newton-Euler method of rigid body dynamics, we can obtain the following equations of motion for the system [4]:

$$\begin{aligned} f_x &= m[\dot{v}_x - \dot{\omega}(\rho_x s_\psi + \rho_y c_\psi) - \omega^2(\rho_x c_\psi - \rho_y s_\psi)] \\ f_y &= m[\dot{v}_y + \dot{\omega}(\rho_x c_\psi - \rho_y s_\psi) - \omega^2(\rho_x s_\psi + \rho_y c_\psi)] \\ \tau_z &= m[-\dot{v}_x(\rho_x s_\psi + \rho_y c_\psi) + \dot{v}_y(\rho_x c_\psi - \rho_y s_\psi)] + J\dot{\omega} \end{aligned} \quad (1)$$

where $c_\psi = \cos \psi$ and $s_\psi = \sin \psi$.

We create a state space model of our dynamic system in the form:

$$\begin{aligned} \dot{x} &= f(x, u, \theta) \\ y &= h(x, u, \theta) \end{aligned} \quad (2)$$

$x = [p_x \ p_y \ \psi \ v_x \ v_y \ \omega]^T$ is the state vector consisting of the x and y inertial frame position coordinates, yaw angle, x and y inertial frame velocities, and z axis angular velocity. u is the control input vector, where each component is the thrust produced by a particular thruster pair aboard the spacecraft. u is related to f_x , f_y , and τ_z by a mixer function that takes into account each thruster's relative position and orientation on the spacecraft, as well as the orientation of the spacecraft itself. $\theta = [m \ \rho_x \ \rho_y \ J_{zz}]^T$ is the vector of mass properties of

interest, consisting of total mass, x and y center of mass offset expressed in the spacecraft body frame, and moment of inertia about the z axis. y is the vector of measurements. We assume that we have full state feedback, so $h(x, u, \theta) = x$.

III. APPROACH

This section details method for generating optimal excitation trajectories. First, we introduce the Fisher Information Matrix, the metric we use to add information weighting to the trajectory generation problem. Next, we formulate the discrete time, finite-horizon optimal control problem. Finally, we present the method for estimating the mass properties based on the excitation trajectory.

A. Fisher Information Matrix

Fisher information is a metric for how much information about a particular quantity is contained within a measurement. The Fisher Information Matrix (FIM) [3] for a given measurement is defined as:

$$F = H^T \Sigma^{-1} H \quad (3)$$

where Σ is the covariance matrix of the measurements, and H is given by:

$$H = \frac{\partial h}{\partial \theta} + \frac{\partial h}{\partial x} \frac{\partial x}{\partial \theta} \quad (4)$$

The partial of the state with respect to the parameters can be computed along a particular trajectory by solving the differential equation with $\Phi = \frac{\partial x}{\partial \theta}$ and $\Phi_0 = 0$:

$$\dot{\Phi} = \frac{\partial f}{\partial x} \Phi + \frac{\partial f}{\partial \theta} \quad (5)$$

For our system, some simplifications can be made. Assuming full state feedback, $\frac{\partial h}{\partial \theta} = 0$, $\frac{\partial h}{\partial x} = I$, and the FIM reduces to $F = \Phi^T \Sigma^{-1} \Phi$.

B. Optimal Excitation Trajectory Generation

We formulate the optimal trajectory generation problem as a finite-horizon discrete optimal control problem:

$$\begin{aligned} \min_{x,u} \quad & J(x, u, \theta) = \sum_{k=0}^h l(x_k, u_k, \theta) \\ \text{s.t} \quad & x_{k+1} = x_k + f(x_k, u_k, \theta) \delta t \\ & u_{min} \leq u_k \leq u_{max} \\ & F_k = \Phi_k \Sigma^{-1} \Phi_k \\ & \Phi_{k+1} = \Phi_k + \left(\frac{\partial f}{\partial x} \Phi_k + \frac{\partial f}{\partial \theta} \right) \delta t \end{aligned} \quad (6)$$

We choose the stage cost [1]:

$$l(x_k, u_k, \theta) = e_k^T Q e_k + u_k^T R u_k + \lambda_k^T \text{diag}(F_k^{-1}) \quad (7)$$

where e is the tracking error from some reference state x_r , and Q , R , and λ are the weights on tracking error, control effort, and information gain respectively. Minimizing the inverse of the FIM is equivalent to maximizing the information gained over the course of the trajectory. Because this is a scalar valued cost function, we need to convert the inverse FIM to

a scalar value. For this work, we choose a linear combination of the diagonal elements of the inverse FIM, allowing us to individually prioritize gathering information about some mass properties over others. Heuristics for choosing the elements of λ_k at each time step are discussed in a later section.

C. Estimation of Mass Properties

We can rearrange equation (1) into a linear equation in terms of the mass properties:

$$A \hat{\theta} = b \quad (8)$$

$$\begin{bmatrix} f_x & \dot{\omega} s_{\psi} + \omega^2 c_{\psi} & \dot{\omega} c_{\psi} - \omega^2 s_{\psi} & 0 \\ f_y & -\dot{\omega} c_{\psi} + \omega^2 s_{\psi} & \dot{\omega} s_{\psi} + \omega^2 c_{\psi} & 0 \\ \tau_z & \dot{v}_x s_{\psi} - \dot{v}_y c_{\psi} & \dot{v}_x c_{\psi} + \dot{v}_y s_{\psi} & -\dot{\omega} \end{bmatrix} \begin{bmatrix} \frac{1}{m} \\ \rho_x \\ \rho_y \\ \frac{j}{m} \end{bmatrix} = \begin{bmatrix} \dot{v}_x \\ \dot{v}_y \\ 0 \end{bmatrix}$$

Given a sequence of measurements, we can assemble an overdetermined linear system:

$$\begin{bmatrix} A_0 \\ \vdots \\ A_k \end{bmatrix} \hat{\theta} = \begin{bmatrix} b_0 \\ \vdots \\ b_k \end{bmatrix} \quad (9)$$

We obtain the best mass properties estimate by finding the least-squares solution of this linear system using the Moore-Penrose pseudoinverse [5].

IV. SIMULATION OVERVIEW

This section describes the simulation framework developed for this project. The core requirements of the simulation are twofold: to generate and solve for information-aware excitation trajectories in a realistic environment, and to allow multiple agents to perform a prescribed transport, stabilization, or system identification mission on an arbitrary unknown superstructure.

The simulation is built on top of MuJoCo [7], and leverages its physics engine to facilitate the contact dynamics and constraint handling necessary to model proper agent-structure docking. We use rigid-body welds to anchor the agents to the superstructure, and MuJoCo then simply couples the joints (which define degrees of freedom) for both bodies, ensuring accurate force and torque transfer specifically at the docking sites.

Each floatbot follows the system model presented above, and uses CasADI [2] with IPOPT to solve the NL OCP within its MPC controller. The structure itself is dynamically generated from block primitives and connection specifications, and can be formed into predefined shapes as well as procedurally generated growths.

A simulation executive oversees superstructure and agent instantiation, and enforces synchronous stepping during run-time. The physics evolve at the default MuJoCo $dt = 0.002$, while the user-defined dt sets the interval for each control step. In each of these, the MuJoCo engine steps forth only once all agents have registered their control actions. Afterwards, the agents then query their state information from a sim-wide Oracle that maintains continuous truth data. If conducting estimation missions, they may choose to update their current set of

superstructure mass estimates during this step as well, which allows for online adaptation of the information parameters.

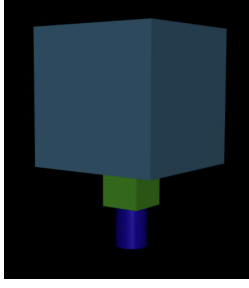


Fig. 1. Prototypical Test Case: Floatbot (dark blue) with attachment voxel (green) and 4 pairs of thrusters (not visible) attached to unit structure block

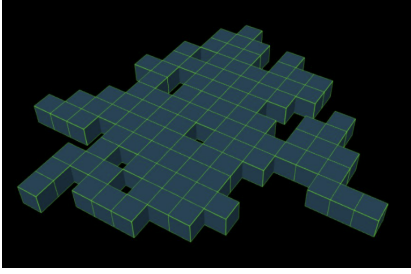


Fig. 2. Sample Procedurally Generated Superstructure, 100 blocks, 1000 kg

V. IMPLEMENTATION

Parameter estimation missions were conducted in simulation on three kinds of structures:

- 1) Unit Block: Single-body, $m = 10kg$
- 2) Ring: 'Hollow' square with blocks only on outer edge
- 3) Random: Randomly generated structure with branches and gaps

On each, the approach in Section 3 was implemented to compute optimal excitation trajectories using the FIM. The results discussed all follow the same (offline plan) $\rightarrow$ (act) $\rightarrow$ (estimate) procedure.

Trajectories are first computed offline by the agent when first added to the structure, at which point it has received a commanded state, and knows only about its own mass properties. This means the dynamics propagation in its cost function constraint uses these currently assumed values. This trajectory is then executed by MuJoCo, where the agent-superstructure assembly evolves using its true compound mass properties. Afterwards, an agent uses its own state history to build the regression matrices and produce a mass properties estimate.

Note that these parameters are *not* updated or recomputed online during the control step and execution yet. The functionality for online evaluation is present (discussed further in Section 8) and will be explored soon, but this report focuses primarily on the computation and execution of these initial trajectories only, and makes conclusions as to the necessity of online parameter adaptation.

VI. RESULTS

Results are presented below. First, we show the impact of the information parameters on the trajectory solutions, followed by a heavier study on information weighting within the cost function. We conclude by considering the parameter adaptation question, which can certainly be addressed with nominal online-MPC style operation, but also by time-dependent exploration scheduling, which we discuss in detail.

A. Optimal Excitation Trajectory

Figures 3 and 4 show the trajectories computed for the simple block test case with and without use of the FIM. These figures show a planned trajectory of a 90 degree rotation of the object. As seen in figure 3, when the FIM is not added to the cost function, the structure completes its assigned rotation but does not move translationally. This prevents the structure from obtaining the optimal measurements to estimate the mass properties. However, in figure 4, when the FIM is used in the cost function, there is more significant translational movement in the x and y directions.

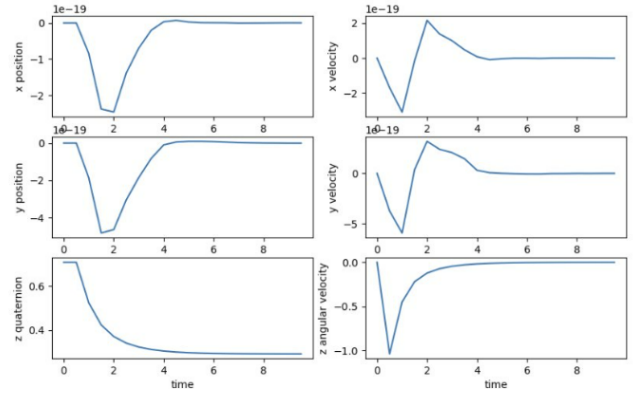


Fig. 3. Optimal Trajectory without Fisher Information Matrix

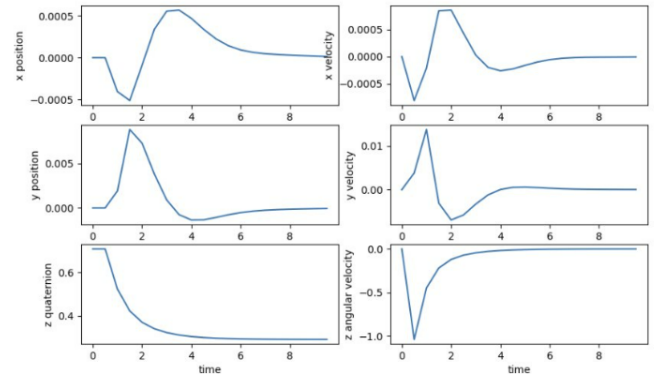


Fig. 4. Optimal Trajectory with Fisher Information Matrix

These two figures show how the Fisher Information Matrix changes the optimal excitation trajectory to add information gathering that will be applied to the mass properties estimation.

B. Parameter Estimation w/ Varying λ

We now conduct a full set of tests on the three structures in which we command and execute a translational trajectory with varying information weighting. We compute the mass estimates and then compare them to the simulation ground truth, and provide preliminary results on information tuning in the cost function. For this set of results, λ is uniform across all mass properties parameters, and Q, R, Σ are fixed at:

$$Q = \text{diag}([10 \ 10 \ 100 \ 10 \ 10 \ 1])$$

$$R = 0.01\mathbf{I}, \quad \Sigma = 0.01\mathbf{I}$$

With appropriate dimensions on $\mathbf{I}$ for R, Σ respectively. λ is logarithmically varied from $\{0.1, 1000\}$ to analyze relative impact against each of the other cost weights.

Mass and inertia estimate errors for the simple block case are presented first, followed by estimates on mass only for the ring and random structures. The inertia estimates for those are extremely poor and not plotted - on the order of 100%–200%, and the reasons why are discussed below.

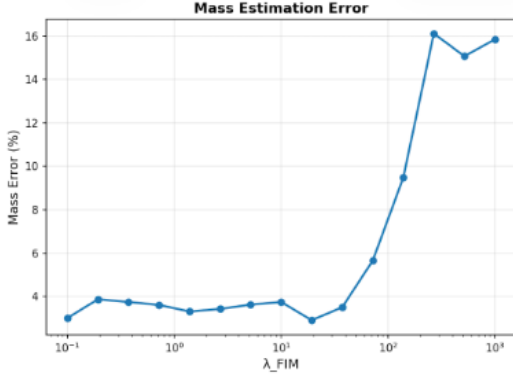


Fig. 5. Unit Block: Mass Estimate Error vs. λ

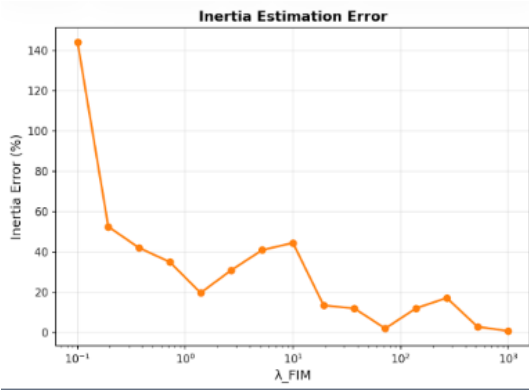


Fig. 6. Unit Block: Inertia Estimate Error vs. λ

Mass estimation for the basic case proved robust, averaging 3 – 4% for most λ , and rising to $\sim 16\%$ at peak λ - which again is weighted evenly for all parameters in this scenario. Inertia estimation is much more λ -dependent, though it does converge to an even better estimate at peak accuracy $\sim 0.8\%$. The improvement in J_{zz} estimate is expected with increasing weight on the λ , but the inverse coupling between that and for

m is of further interest. An optimal operating region exists roughly at $\lambda \sim 100$, though this is still somewhat poorly conditioned and on the threshold of rising inaccuracy in the mass estimate.

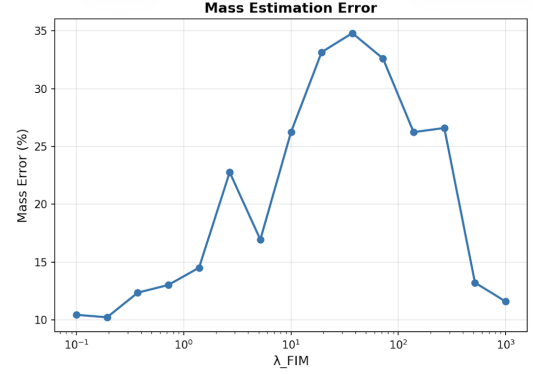


Fig. 7. Ring: Mass Estimate Error vs. λ

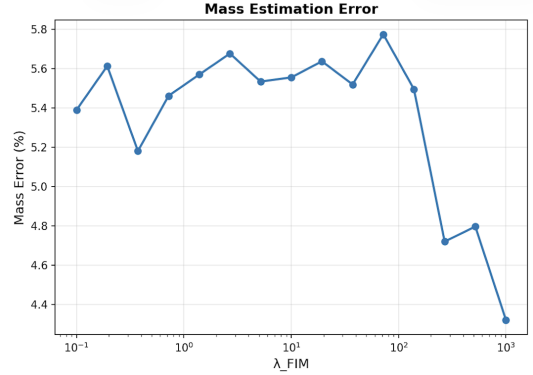


Fig. 8. Random Structure: Mass Estimate Error vs. λ

The mass estimates for the ring scenario are much poorer, though surprisingly the random structure performs the best of the set. The ring case is being investigated further to determine how the parameter weights impact the final inertia estimate, and how heavily this influences the mass calculation. We note that such a structure will have a significantly higher J_{zz} due to its mass concentration, and that this might be much harder for the agent to estimate. These estimates are computed after the entire trajectory has been executed, so the initial mass parameters assumption of the agent (just itself) is bound to have enormous error in J_{zz} , which compounds over execution.

In all cases, the inertia estimate is worst at lowest λ , which is expected. The commanded trajectory is a translation; low λ means no rotation is added to the trajectory, and less information is gathered about the inertia, while the mass estimates benefit from the default movement. Essentially, this creates an poorly conditioned least squares estimate in the regression, which improves with increased information weight. The previous discussion about initial inertia offsets amplifying through the trajectory execution - due to lack of online update - also applies here, and the inertia offsets were expectedly higher in the ring vs. random scenario.

C. Non-Uniform λ

We now present preliminary results on non-uniform parameter weighting in the FIM. In this set of cases, conducted on the unit block, $\lambda_{CG} = 100$, Q, R, Σ remain the same, but $\lambda_{mass}, \lambda_{J_{zz}}$ are each varied over the same range as the uniform λ was before. Regions with error over 30% are clipped for clarity and to reveal the sensitivity of the estimates with changes in individual parameter weights.

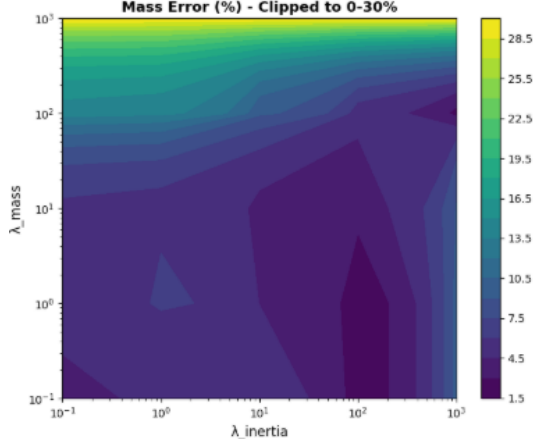


Fig. 9. Mass Estimate as $f(\lambda_{mass}, \lambda_{J_{zz}})$

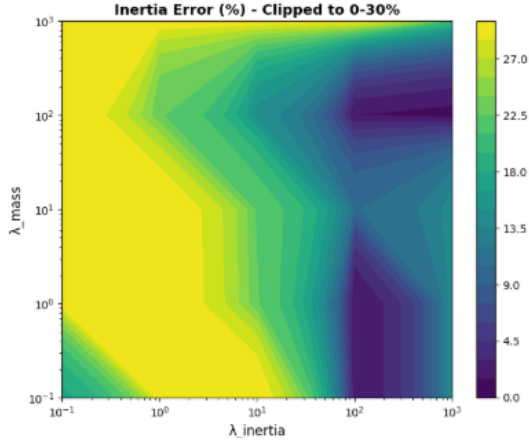


Fig. 10. Inertia Estimate as $f(\lambda_{mass}, \lambda_{J_{zz}})$

As expected, being able to non-uniformly vary the weights can lead to much better overall estimation for all parameters. For this translational case, prioritizing inertial information over the already guaranteed mass data leads to an improved inertial estimate. Furthermore, the optimal performance of this controller actually approaches $< 1\%$ estimation accuracy for both mass and inertia, much better than the uniform scenario.

We have discussed the fact that this implementation is offline and that estimation happens only after execution. Allowing the agents to update their stored estimate at each control step will certainly improve accuracy, especially for longer, multi-waypoint or multi-agent estimation missions. However, we are also interested in ways to implement adaptive λ schemes that can plan a trajectory assuming improvement in knowledge over time. These are discussed next.

D. Information with Exploration Scheduling

Since the purpose of adding an information term is to determine the mass properties with a deliberately exploratory maneuver, it is then desirable to dynamically shift towards a purely goal state oriented trajectory once the mass properties have been adequately estimated. We implement this time-dependent adaptive λ schedule using the following predictive heuristic for altering the weight on the FIM,

$$\lambda_k = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min})e^{-\gamma k} \quad (10)$$

where the weight, λ_k starts at its initial maximum bound and decreases by a decay factor, γ , towards its minimum bound for each discrete timestep, k . For this demonstration, changing λ was treated as a scalar function and so the cost function acts on the trace of the FIM [1].

$$J(x, u, \theta) = \sum_{k=0}^h e_k^T Q e_k + u_k^T R u_k + \lambda_i^T \text{Tr}(F_i^{-1}) \quad (11)$$

This changing of λ can be demonstrated by comparing two oppositely tuned decay rates, with the following trajectory states plotted against a 10 second time horizon, a minimum λ of 1, maximum λ of 500, and a decay factor of $\gamma = 0.2$.

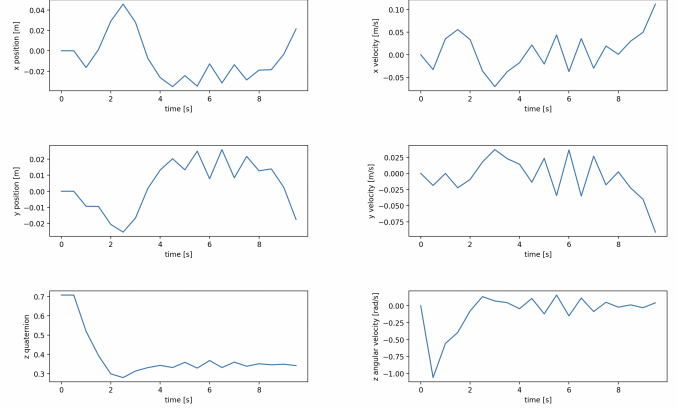


Fig. 11. Optimal Trajectory with decay rate of 0.2

Using a decay factor that is closer to 0 than to 1 enforces a schedule that effectively transitions from high information-gathering priority to low information-gathering priority, more rapidly than a set that uses a higher decay factor. This implies that the designer predicted that a longer timescale is needed for the measured mass properties to converge. Conversely, the opposite effect can be shown with a decay factor of $\gamma = 0.9$.

The oscillatory motion shown later in the time horizon is characteristic of the information term still having a high relative weight on the total cost function. As expected, the opposite tuning approach of the $\gamma = 0.9$ schedule shows the opposite effect, where the smooth trajectory is designed to place a higher priority on the state later in the time horizon, since the assumption is that the mass properties will have converged early. One way to test the usefulness of a chosen decay factor is by how quickly the mass properties converge

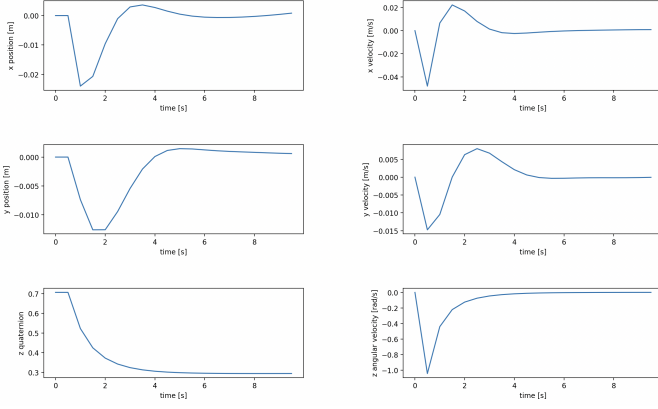


Fig. 12. Optimal Trajectory with decay rate of 0.9

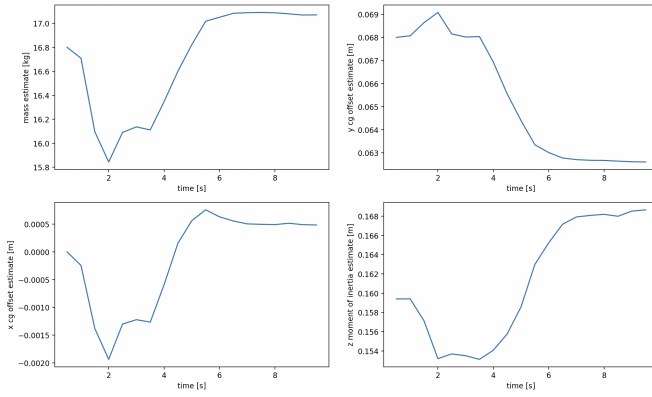


Fig. 13. Mass convergence demonstrated for $\gamma = 0.2$

by inspection. In the $\gamma = 0.2$ case, we see the following for the 4 mass properties.

The mass properties begin to hover around a convergence value (though not adequately stable) at approximately 6 seconds, indicating that the oscillating maneuvers (which suggest a high exploratory control priority) later in the trajectory (>6 seconds) could be unnecessary. It implies an excessive control effort and a state error at that point in time for the same results. This then leads to the mass convergence evaluation for the $\gamma = 0.9$ schedule.

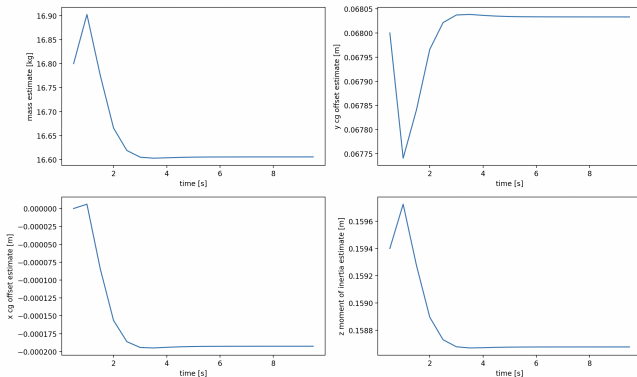


Fig. 14. Mass convergence demonstrated for $\gamma = 0.9$

The first observation is that it seemingly converges better despite putting a lower priority on information collection than its counterpart in this example. It could then be confidently suggested that in this particular case, a faster decay, like the $\gamma = 0.9$ schedule, is the optimal choice from pure inspection. However, though quick convergence is attained in this case, the mass stabilizes around 0.2 kg below the true value, meaning the mass was measured with this error, and then the lack of exploration priority likely prevented it from correcting itself.

The data suggest that both cases fall short of the optimum scheduling. Further manual tuning suggests potential improvements, but it is not clear how worthwhile they would be given other variables like the state's initial conditions and the range of λ . Though the above process seems like an intuitive approach, it would be more desirable, especially if the states and dynamics grow complex enough to where this tuning λ by inspection becomes infeasible anyway, to have a system where the weight on the FIM is dynamically dependent on the sensor measurements. This would require a quantification of the uncertainty of real time measurements with a scheme appropriately tailored to a specific type of sensor data. This paper seeks to propose future directions for this topic in the succeeding sections.

VII. FUTURE WORK

Presently, we have demonstrated results on information-aware excitation trajectories and associated controller information weight tuning, for single-agent scenarios across three nominal structures. This section briefly describes our intended roadmap to full, coordinated, multi-agent parameter estimation and trajectory generation.

A. Online Estimation and Test Expansion

Immediate steps are to toggle online estimation on for the agents, allowing them to update their mass properties at each control step. This will be accompanied by an expanded test suite that critically features different commanded trajectories, rather than the pure translation results discussed here. Results here will augment this report in demonstrating the need for online parameter adaptation, and results of interest will be comparisons between this online method against the λ -scheduling.

B. Cost Function Tuning

A preliminary grid sweep over uniform and non-uniform λ has been conducted, but we will soon expand this to a more robust sampling-based approach or random search over this space, ideally followed by some sort of optimization scheme. This would lead to a Pareto-style analysis where we are able to evaluate trade-offs in estimation errors for all the parameters.

C. Multi-Agent Expansion

Currently, the simulation supports N agents (limited by compute), and each can possess a unique system model, controller, and reference state. Communication and information between the agents is not yet implemented, so any coordinated

missions are completely decentralized. An underlying graph representation of the multi-agent fleet is being developed in NetworkX, which will allow us to govern inter-agent information exchange by rules we determine on the network topology, such as those on distance, occlusion, line of sight, etc. Rather than communicating directly, agents will submit queries to the Oracle for information about each other. The Oracle will then evaluate these requests against the current rule set, and work with a coordination manager in order to grant or deny these queries accordingly.

We would like to tune the degree of decentralization and coordination by varying the weighting between local vs. global cost functions where possible over the following spectrum:

- 1) Centralized: Global entity solves complete OCP and simply delegates tasks
- 2) Ego-Centric: Each agent assumes it is the only actor - perhaps on a structure with mass normalized by number of agents
- 3) ZOH/Tension: Each agent assumes all other agents will command a fixed control action or constant holding force
- 4) Info Exchange: Agents begin with assumptions similar to (2), (3) but are now allowed to exchange info on states, actions, and guarantees on these with each other, and adjust their plans accordingly
- 5) Auto-Decomposed: All agents receive a global goal, and the network autonomously *decomposes* it into heterogeneous individual cost functions, determined dynamically by the swarm topology, deviation from goal state, and confidence in the information content

Of these, we anticipate that a centralized approach should uniformly provide better mass estimates and swarm control authority, at the obvious expense of responsiveness. We are also interested in seeing how strategies aligned with (2,3) perform - some approaches for decentralized swarm stabilization of passive structures have shown remarkable success given the lack of info-sharing; but we hypothesize that actions from other agents in our scenario may be far more adversarial in terms of their ability to corrupt a neighbor's local excitation trajectory $\rightarrow$ mass property estimation computation. Results from (4) will be of chief interest - for now, the topology of the swarm will only prohibit messages between agent pairs that are out-of-range or visually occluded, and we will have a set of deterministic communications protocols that can tune the amount of information shared. (5) is likely out of scope for this project, but begins to address ideas related to self-organization and autonomous, dynamic establishment of hierarchies within a network.

D. Sensor Integration

There is no state estimation or localization occurring in this simulation at an agent level; they receive their true state from the sim at every step. MuJoCo offers a suite of IMU, contact, and vision sensors that can be integrated into the agent model, and this sim will eventually integrate these as it moves to full ROS2-support in preparation of a hardware demo - but for the purposes of this project, they receive perfect state.

This does open a desire to explore both the measurement methods on-orbit and how the uncertainty might be quantified. A study of on-orbit target mass property measurements was conducted that outlined the use of pose-graph mapping to transform remote sensing data into mass property data, specifically the center of mass [Map]. It is worth investigating this mapping method to determine if the uncertainty from this technique [Map] could reduce, and thus dynamically tune λ as an answer to the heuristic use of time in the decaying λ case.

VIII. CONCLUSION

This report has presented a method for estimating unknown mass properties of an arbitrary unknown structure, determined by an agent executing an information-aware excitation trajectory using a Fisher Information Matrix. We test this method in a high-fidelity simulation engine on three nominal test structures, and analyze the accuracy of this parameter estimation across various controller configurations and information weights. Results demonstrate how trajectories are modified to gather more information about the unknown parameters, and the sweep on parameter weights indicates both the utility in non-uniform λ , but also the sensitivity of the estimate to these terms. Ultimately, these results highlight the need for online adaptation of the FIM with periodic parameter updates, and also present some alternative methods for scheduling λ beforehand. We finish with a brief discussion of current limitations and the preliminary requirements for a multi-agent implementation of these methods.

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